



General

- Fast Acting
- Surface mount high current fuse
- Higher voltage rating up to 125VDC
- 12.1mm×10.0mm×6.8mm square shape surface mount
- -55°C to 125°C Operating temperature
- Excellent environmental integrity
- Enhanced thermal cycling endurance
- RoHS compliant
- Halogen Free
- Pb Free

Agency / Certificate Information

Agency	File Number	Ampere Range
	E527955	60A~150A
	J50657756	60A~150A

Application

- Storage system power
- Cooling fan system for PC server
- Voltage regulator module
- Base station power supply
- Voltage regulator module for PC server
- High end servers / Blade computing
- Battery Management System

Electrical Specifications

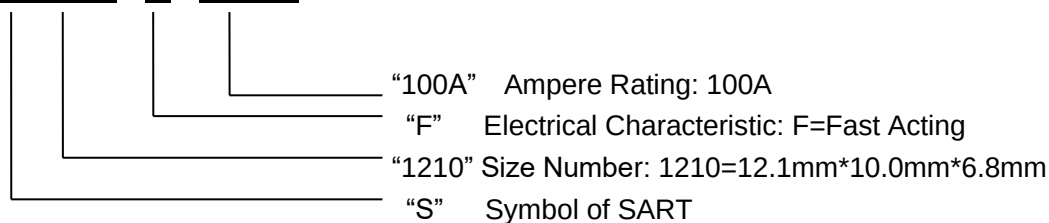
Part Number	Current Rating (A)	Voltage Rating (VDC)	Interrupting Rating		Typical Cold DCR* (mΩ)	Nominal I ² T** (A ² sec)
S1210-F-60A	60	125	UL 1000A @ 125VDC 5000A @ 100VDC	TUV 2000A @ 125VDC 5000A @ 100VDC	0.98	2100
S1210-F-70A	70	125			0.75	2800
S1210-F-80A	80	125			0.57	3400
S1210-F-90A	90	125			0.52	4000
S1210-F-100A	100	125			0.46	4300
S1210-F-125A	125	125		TUV 1000A @ 125VDC 5000A @ 100VDC	0.33	6250
S1210-F-150A	150	125			0.29	11250

* Measured at ≤10% rated current and 25°C

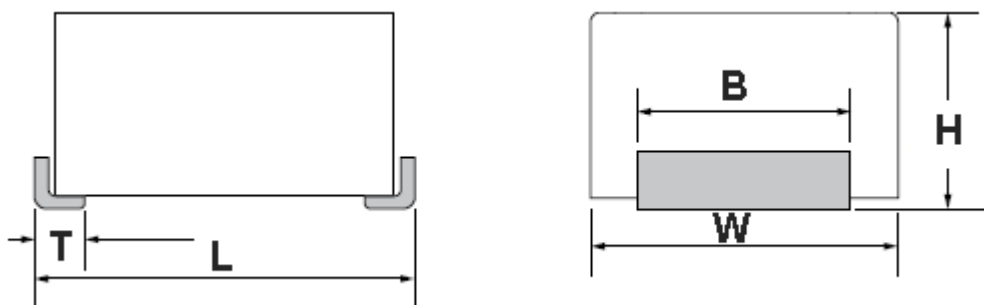
** Melting I²T at 10 times of rated current

Part Number Information

S1210-F-100A

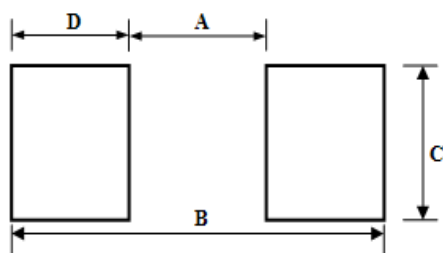


Dimensions



Type	L (mm)	W (mm)	H (mm)	T (mm)	B (mm)
S1210-F	12.00±0.50	10.00±0.30	6.50±0.30	1.80±0.30	6.30±0.30

Recommended Land Patterns

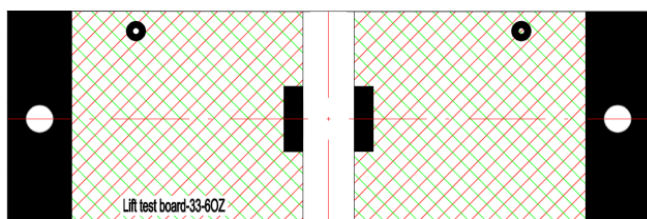


Materials

Components	Material
Body	Ceramic
Element	Tin Plated Copper

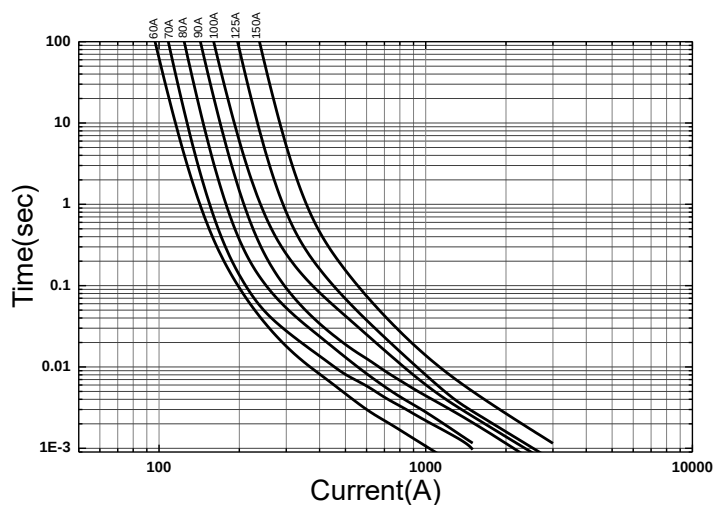
Type	A(mm)	B (mm)	C(mm)	D(mm)
S1210-F	8.00±0.30	14.00±0.30	10.00±0.30	3.00±0.30

Dimensions of Standard Test Board

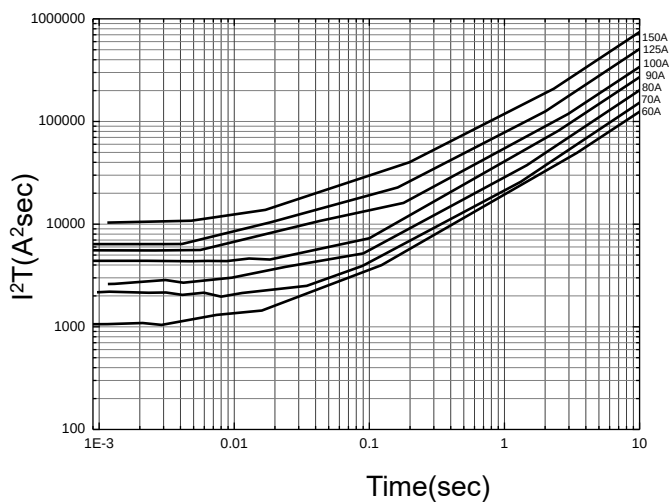


Board thickness: 1.6mm; Thickness of Copper layer: 280um+5um;
 Width of Copper trace: 33mm

Time Current Curve



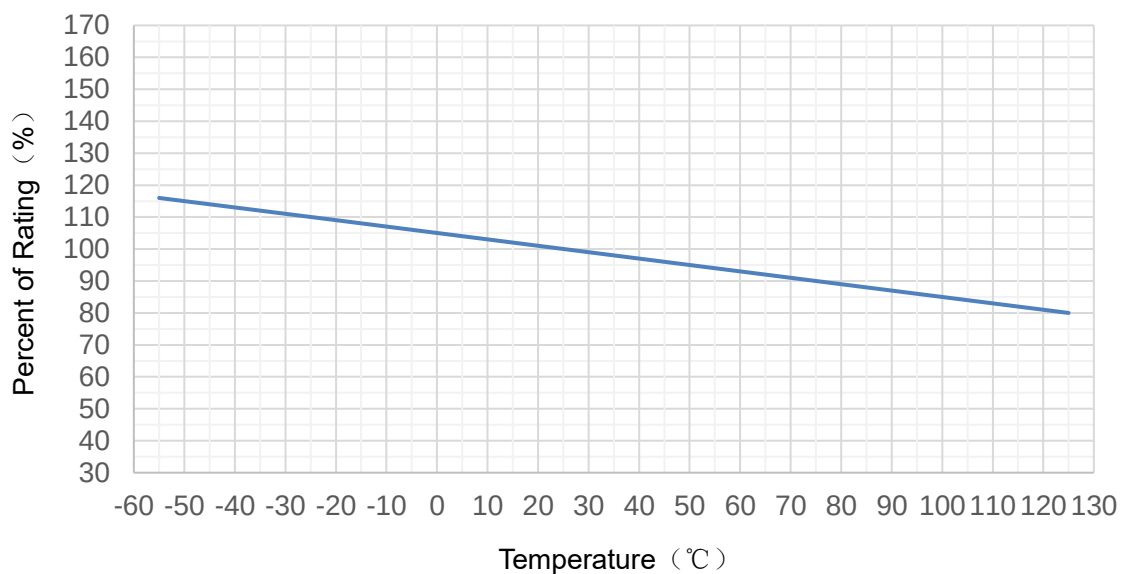
I²T VS Time Curve



Electrical Characteristics

Type	Ampere Rating	% of Current Rating	Opening Time
S1210-F	60A~150A	100%	Min.4hrs
	60A~150A	200%	Max.60sec
	60A~150A	1000%	Max.10msec

Temperature Derating Curve



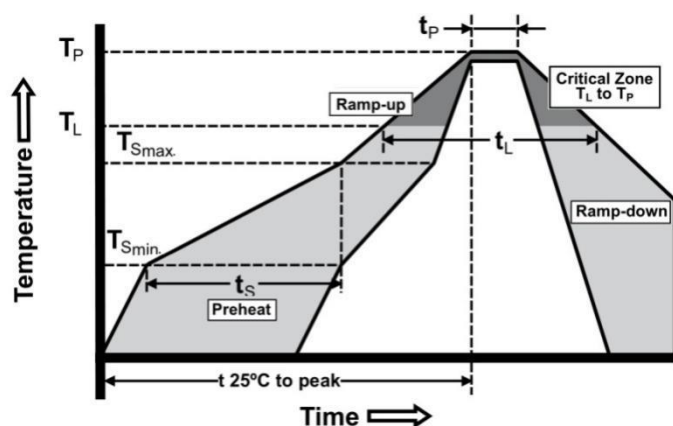
Product Characteristics

Item	Test condition/ Methods	Performance	Standard
Time/Current	100% of current rating	No Fusing; 4hrs Min.	UL248-14
	200% of current rating	60sec Max.	SART SPEC
	1000%of current rating	10msec Max.	SART SPEC
Interrupting Ability	60A~150A (UL / TUV) : 100VDC 5000A 60A~150A (UL) : 125VDC 1000A 60A~100A (TUV) : 125VDC 2000A 125A~150A (TUV) : 125VDC 1000A	without permanent arcing, ignition and bursting of fuse link	UL248-14 IEC60127-7
Solderability	240℃±5℃, 3sec±0.5sec	95% coverage Min.	IEC60127-4 IEC60068-2-20; MIL-STD-202
Resistance to soldering	260℃±5℃, 10sec±0.5sec	ΔR : <10%	MIL-STD-202 Method 210
Moisture resistance	Temperature Humidity, T: 85℃±3℃, RH: 85%±5%, Duration: 1000 hrs	ΔR : <10%	MIL-STD-202 Method 106
Insulation Resistance	Test Condition A, 0.01MΩ Min.	0.01MΩ Min.	Mil-STD-202 Method 302
Low Temperature Storage	T=-55℃±3℃, 1000hrs	ΔR : <10%	IEC60068-2-1
High Temperature Storage	T=125℃±2℃, 1000hrs	ΔR : <10%	IEC60068-2-2
Salt Spray	5%±1% salt solution, 72hrs	ΔR : <10%	MIL-STD-202 Method 101
Moisture Sensitivity Level	Level 1	ΔR : <10%	J-STD-020
Thermal Shock	100 cycles, -65℃ to +125℃, 30 minutes @ each extreme	ΔR : <10%	IEC 60068-2-14
Vibration	Amplitude 10Hz~55Hz in 1 min. 2 hrs each XYZ, total 6 hrs	ΔR : <10%	MIL-STD-202F Method 201
Mechanical shock	100g peak amplitude, saw tooth wave, 6msec duration, 3 cycles XYZ+xyz = 18	ΔR : <10%	MIL-STD-202 Method 213

Recommended Solder Curve

1. Infrared Reflow:

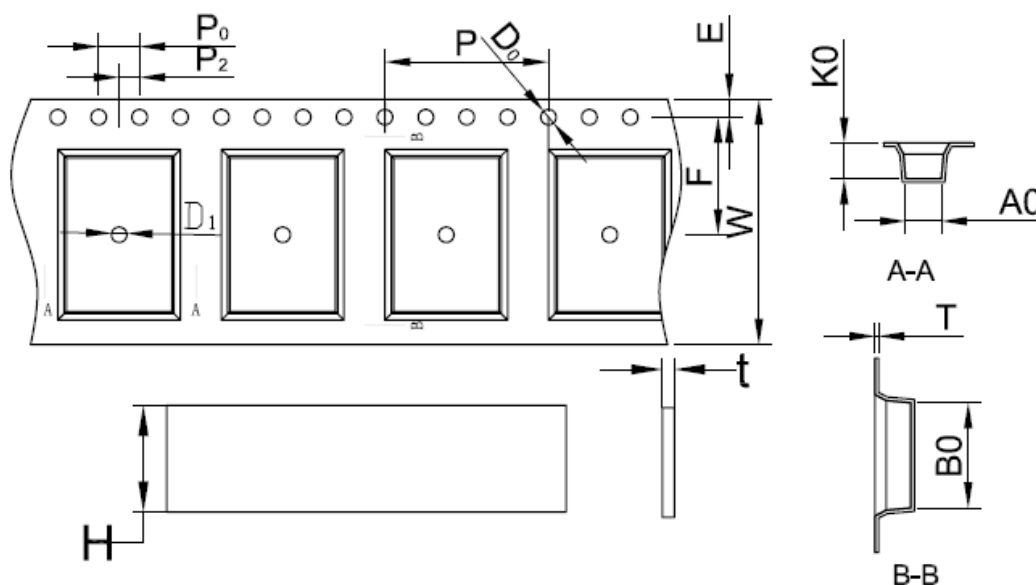
- Temperature: 260°C
- Time: 20sec Max.
- Recommend Reflow profile



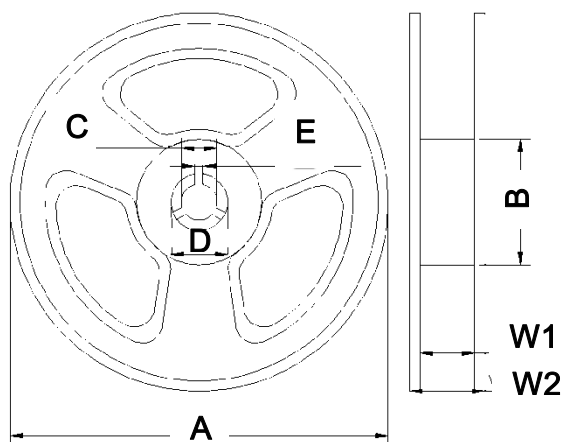
Profile Feature	Pb-Free Assembly
Average Ramp-Up Rate(T_{Smax} to T_P)	3°C/sec Max.
Preheat Temperature Min(T_{Smin}) Temperature Max(T_{Smax}) Time(T_{Smin} to T_{Smax})	150°C 200°C 60sec ~ 120sec
Peak Temperature(T_P)	260°C
Time within 5°C of actual Peak Temperature(t_P)	20sec
Temperature(T_L)	217°C
Melting tin time(t_L)	60sec ~ 150sec
Ramp-Down Rate	6°C/sec Max.
Time 25°C to Peak Temperature	8 min Max.

Packaging

500 pieces of fuses on 24mm tape-and-reel on 13 inch (330mm) reel



Type	A0(mm)	B0(mm)	K0(mm)	P0(mm)	P(mm)	P2(mm)	E(mm)
S1210-F	10.30±0.10	13.00±0.15	7.20±0.10	4.00±0.10	16.00±0.10	2.00±0.10	1.75±0.10
Type	F(mm)	D0(mm)	D1(mm)	W(mm)	T(mm)	H(mm)	t(mm)
S1210-F	11.50±0.10	1.50 ^{+0.10} ₋₀	1.50 ^{+0.10} ₋₀	24.00±0.30	0.35±0.05	21.30±0.15	0.055 ^{+0.015} _{-0.007}



Type	A(mm)	B(mm)	C(mm)	D(mm)	E(mm)	W1(mm)	W2(mm)
S1210-F	330.0±2.0	100.0±1.5	13.0±0.5	21.0±0.5	2.2±0.2	24.5±1.5	28.5±2.0

Storage

- The ambient temperature recommended for storage shall be between 5°C~30°C.
- The relative humidity recommended for storage shall be between 25%RH~60%RH.
- Sealed plastic bags with desiccant shall be used to reduce the oxidation of the termination and shall only be opened prior to use.
- The products shall not be stored in areas where harmful gases containing sulfur or chlorine are present.